Course Title: Cryptography and Network Security Lab	**Course Code**: BCSE309P
Faculty: Prof. Karthika V	Slot: L15+L16
Regno: *23BCE1344*	Name: *Aditya*

EX1C

TRANSPOSITION CRYPTOGRAPHIC ALGORITHM

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Code: (--CyptoFunctions--)

Cypto_functions.py:

```
# crypto_functions.py
import math

def rail_fence_encrypt(text, key):
    rails = ['' for _ in range(key)]
    row, direction = 0, 1
    for ch in text:
        rails[row] += ch
        if row == 0:
            direction = 1
        elif row == key - 1:
            direction = -1
        row += direction
    return ''.join(rails)

def rail_fence_decrypt(cipher, key):
    rail = [['\n' for _ in range(len(cipher))] for _ in range(key)]
    row, direction = 0, 1
    for col in range(len(cipher)):
        rail[row][col] = '*'
        if row == 0:
            direction = 1
        elif row == key - 1:
            direction = -1
        row += direction
    index = 0
    for i in range(key):
        for j in range(len(cipher)):
            if rail[i][j] == '*' and index < len(cipher):
                rail[i][j] = cipher[index]
                index += 1
    result = []
    row, direction = 0, 1
    for col in range(len(cipher)):
        result.append(rail[row][col])
        if row == 0:
            direction = 1
        elif row == key - 1:
            direction = -1
        row += direction
    return ''.join(result)

def row_column_encrypt(text, key):
    text = text.replace(" ", "")
    key = [int(k) for k in key]
    cols = len(key)
    rows = math.ceil(len(text) / cols)
    text += '*' * (rows * cols - len(text))
    matrix = [text[i:i+cols] for i in range(0, len(text), cols)]
    order = sorted(range(cols), key=lambda i: key[i])
    cipher = ""
    for col in order:
```

```python
        for row in matrix:
            cipher += row[col]
    return cipher

def row_column_decrypt(cipher, key):
    key = [int(k) for k in key]
    cols = len(key)
    rows = math.ceil(len(cipher) / cols)
    matrix = [['' for _ in range(cols)] for _ in range(rows)]
    order = sorted(range(cols), key=lambda i: key[i])
    index = 0
    for col in order:
        for row in range(rows):
            matrix[row][col] = cipher[index]
            index += 1
    plain = ""
    for row in matrix:
        plain += ''.join(row)
    return plain.rstrip('*')

# MAIN FUNCTION (TESTING)
def main():
    print("RAIL FENCE CIPHER")
    msg = "HELLO WORLD"
    key = 3
    enc = rail_fence_encrypt(msg, key)
    dec = rail_fence_decrypt(enc, key)
    print("PT :", msg)
    print("CT :", enc)
    print("DT :", dec)
    print()
    print("ROW COLUMN CIPHER")
    msg = "attack postponed until two am"
    key = "4312567"
    enc = row_column_encrypt(msg, key)
    dec = row_column_decrypt(enc, key)
    print("PT :", msg)
    print("Key:", key)
    print("CT :", enc)
    print("DT :", dec)

if __name__ == "__main__":
    main()
```

Output:

```
(lab) ak@DevLab:~/Desktop/Data/College/Semester6/Assignment/Crypto Lab/Ex1c$ python ./crypto_functions.py
RAIL FENCE CIPHER
PT : HELLO WORLD
CT : HOREL OLLWD
DT : HELLO WORLD

ROW COLUMN CIPHER
PT : attack postponed until two am
Key: 4312567
CT : ttnaaptmtsuoaodwcoi*knl*pet*
DT : attackpostponeduntiltwoam
```

Code: (--Rail Fence Cipher--)

Sender (server.py):

```
import socket
import crypto_functions as cf

key = 3
def encrypt(text):
   return cf.rail_fence_encrypt(text,key)
def decrypt(text):
   return cf.rail_fence_decrypt(text,key)
server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

host = "127.0.0.1"
port = 12345
server_socket.bind((host,port))
server_socket.listen(1)
print(f"Server listening on {host}:{port}")
connection, address = server_socket.accept()
print(f"Connected to {address}")

while True:
   data = connection.recv(1024).decode()
   print(f"Encrypted message from client: {data}")
   data = decrypt(data)
   print(f"Decrypted message from client: {data}\n")
   if data.lower() == "exit":
       print("Exiting...")
       break
   msg = input("You: ")
   enc_msg = encrypt(msg)
   print(f"Encrypted message sent to client: {enc_msg}\n")
   connection.send(enc_msg.encode())
connection.close()
server_socket.close()
```

Receiver (client.py):

```
import socket
import crypto_functions as cf

key = 3
def encrypt(text):
   return cf.rail_fence_encrypt(text,key)
def decrypt(text):
   return cf.rail_fence_decrypt(text,key)
client_socket = socket.socket(socket.AF_INET,socket.SOCK_STREAM)

host = "127.0.0.1"
port = 12345
client_socket.connect((host, port))
print("Conected to the server")
while True:
   message = input("You: ")
   message = encrypt(message)
   print(f"Encrypted message sent to server: {message}\n")
   client_socket.send(message.encode())
   if message.lower() == "exit":
       break
   msg = client_socket.recv(1024).decode()
   print(f"Encrypted message from server: {msg}")
   data = decrypt( msg)
   print(f"Decrypted message from server: {data}\n")
client_socket.close()
```

Output:

```
(lab) ak@DevLab:~/Desktop/Data/College/Semester6/Assignment/Crypto La
b/Ex1c$ python server.py
Server listening on 127.0.0.1:12345
Connected to ('127.0.0.1', 42652)
Encrypted message from client: Hoell
Decrypted message from client: Hello

You: meet me after the toga party
Encrypted message sent to client: m aregaetm fe h oapryeettt t
```

```
(lab) ak@DevLab:~/Desktop/Data/College/Semester6/Assignment/Cr
b/Ex1c$ python client.py
Conected to the server
You: Hello
Encrypted message sent to server: Hoell

Encrypted message from server: m aregaetm fe h oapryeettt t
Decrypted message from server: meet me after the toga party

You:
```

Code: (--Vigenere Cipher--)

Sender (server.py):

```
import socket
import crypto_functions as cf

key = "4312567"
def encrypt(text):
   return cf.row_column_encrypt(text,key)
def decrypt(text):
   return cf.row_column_decrypt(text,key)
server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

host = "127.0.0.1"
port = 12345
server_socket.bind((host,port))
server_socket.listen(1)
print(f"Server listening on {host}:{port}")
connection, address = server_socket.accept()
print(f"Connected to {address}")

while True:
   data = connection.recv(1024).decode()
   print(f"Encrypted message from client: {data}")
   data = decrypt(data)
   print(f"Decrypted message from client: {data}\n")
   if data.lower() == "exit":
       print("Exiting...")
       break
   msg = input("You: ")
   enc_msg = encrypt(msg)
   print(f"Encrypted message sent to client: {enc_msg}\n")
   connection.send(enc_msg.encode())
connection.close()
server_socket.close()
```

Receiver (client.py):

```
import socket
import crypto_functions as cf

key = "4312567"
def encrypt(text):
   return cf.row_column_encrypt(text,key)
def decrypt(text):
   return cf.row_column_decrypt(text,key)
client_socket = socket.socket(socket.AF_INET,socket.SOCK_STREAM)

host = "127.0.0.1"
port = 12345
client_socket.connect((host, port))
print("Conected to the server")
while True:
   message = input("You: ")
   message = encrypt(message)
   print(f"Encrypted message sent to server: {message}\n")
   client_socket.send(message.encode())
   if message.lower() == "exit":
       break
   msg = client_socket.recv(1024).decode()
   print(f"Encrypted message from server: {msg}")
   data = decrypt( msg)
   print(f"Decrypted message from server: {data}\n")
client_socket.close()
```

Output:

```
(lab) ak@DevLab:~/Desktop/Data/College/Semester6/Assi
gnment/Crypto Lab/Ex1c$ python server.py
Server listening on 127.0.0.1:12345
Connected to ('127.0.0.1', 32896)
Encrypted message from client: lleHo**
Decrypted message from client: Hello

You: attack postponed until two am
Encrypted message sent to client: ttnaaptmtsuoaodwcoi
*knl*pet*
```

```
(lab) ak@DevLab:~/Desktop/Data/College/Semester6/Assi
gnment/Crypto Lab/Ex1c$ python client.py
Conected to the server
You: Hello
Encrypted message sent to server: lleHo**

Encrypted message from server: ttnaaptmtsuoaodwcoi*kn
l*pet*
Decrypted message from server: attackpostponeduntiltw
oam

You:
```

python
pyth...
bash